

Homework (Habanero)

- Q1.** Simplify the expression $\sqrt{16} + \sqrt{9}$
(A) 7
(B) 5
(C) 4
(D) 3
(E) 1
- Q2.** What is the product of $\sqrt{12}$ and $\sqrt{3}$?
(A) $\sqrt{36}$
(B) $\sqrt{20}$
(C) $\sqrt{15}$
(D) $\sqrt{12}$
(E) $\sqrt{9}$
- Q3.** A patient is prescribed 250mg of a medication. If the dosage is increased by $\sqrt{2}$ times, what is the new dosage?
(A) 350mg
(B) 375mg
(C) 400mg
(D) 500mg
(E) 353.55mg
- Q4.** The rate of cell growth is modeled by $\sqrt{t}$, where t is time in hours. If $t = 16$, what is the rate of cell growth?
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32
- Q5.** Simplify the expression $\frac{\sqrt{24}}{\sqrt{6}}$
(A) $\sqrt{2}$
(B) $\sqrt{4}$
(C) $\sqrt{6}$
(D) 2
(E) $\sqrt{3}$
- Q6.** A statistical analysis of patient data yields a standard deviation of $\sqrt{20}$. What is the variance?
(A) 20
(B) 10
(C) 5
(D) 4
(E) 40
- Q7.** What is the value of $\sqrt{48}$?
(A) $4\sqrt{3}$
(B) $2\sqrt{6}$
(C) $6\sqrt{2}$
(D) $3\sqrt{4}$
(E) $\sqrt{24}$
- Q8.** A medication's half-life is $\sqrt{2}$ hours. If the initial amount is 1000mg, how much is left after $\sqrt{2}$ hours?
(A) 500mg
(B) 707.11mg
(C) 353.55mg
(D) 250mg
(E) 100mg

Open Ended Questions (FRQ)

- Q9.** A certain cell doubles in size every 2 hours. If the initial size is $\sqrt{4}$ cm, how many hours will it take to reach $\sqrt{16}$ cm in size? Show your work and calculations.
- Q10.** A pharmaceutical company produces medication in doses of $\sqrt{25}$ mg. If they want to create a new dose that is 1.5 times the current dose, what is the new dose? Calculate the new dose using radicals and simplify your answer.

Chef's Tips

- Provide clear instructions and examples for each problem type.
- Ensure students understand the context of word problems.
- Remind students to simplify their radical expressions.